

Sample Questions for CO1 – AT1 – Data Interpretation

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10.py - C:/query processing/10.py (3.14.0)
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# Employee data
data = [
    {"EmpID": 101, "Name": "John", "Department": "IT", "Salary": 65000},
    {"EmpID": 102, "Name": "Ram", "Department": "HR", "Salary": 55000},
    {"EmpID": 103, "Name": "Priya", "Department": "IT", "Salary": 72000},
    {"EmpID": 104, "Name": "Anu", "Department": "Finance", "Salary": 58000},
    {"EmpID": 105, "Name": "Ravi", "Department": "IT", "Salary": 61000}
]

# Create DataFrame
df = pd.DataFrame(data)

# Employees with Salary > 60000
print("Employees with Salary > ₹60,000")
print(df[df["Salary"] > 60000])

# Employees in IT Department
print("\nEmployees in IT Department")
print(df[df["Department"] == "IT"])
```

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IDLE Shell 3.14.0
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6 July 42000
7 August 39000
8 September 41000
9 October 45000
10 November 43000
11 December 47000

Total Sales: 453000
Average Sales: 37750.0
Maximum Sales: 47000
Minimum Sales: 25000

>>>
===== RESTART: C:/query processing/10.py =====
Employee Details:
{'John': 50000, 'Ram': 45000, 'Priya': 65000, 'Anu': 55000, 'Ravi': 40000}

Highest Salary Employee: Priya - 65000
Lowest Salary Employee: Ravi - 40000
Average Salary: 51000.0

>>>
===== RESTART: C:/query processing/10.py =====
Marks: [75, 82, 91, 45, 67, 58, 39, 88, 95, 72]
Highest Mark: 95
Lowest Mark: 39
Average Mark: 71.2
Number of Students Passed: 8

>>>
===== RESTART: C:/query processing/10.py =====
Employees with Salary > ₹60,000
  EmpID  Name Department  Salary
0    101   John         IT   65000
2    103  Priya         IT   72000
4    105   Ravi         IT   61000

Employees in IT Department
  EmpID  Name Department  Salary
0    101   John         IT   65000
2    103  Priya         IT   72000
4    105   Ravi         IT   61000

>>>
```

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10.py - C:/query processing/10.py (3.14.0)
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marks = [75, 82, 91, 45, 67, 58, 39, 88, 95, 72]

print("Marks:", marks)

print("Highest Mark:", max(marks))
print("Lowest Mark:", min(marks))
print("Average Mark:", sum(marks)/len(marks))

passed = 0
for mark in marks:
    if mark >= 50:
        passed += 1

print("Number of Students Passed:", passed)
```

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7 Mobile 600 51000
8 SSD 80 4800
9 Camera 450 38250

>>>
===== RESTART: C:/query processing/10.py =====
Monthly Sales:
  Month  Sales
0  January 25000
1  February 30000
2  March 28000
3  April 35000
4  May 40000
5  June 38000
6  July 42000
7  August 39000
8  September 41000
9  October 45000
10 November 43000
11 December 47000

Total Sales: 453000
Average Sales: 37750.0
Maximum Sales: 47000
Minimum Sales: 25000

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===== RESTART: C:/query processing/10.py =====
Employee Details:
{'John': 50000, 'Ram': 45000, 'Priya': 65000, 'Anu': 55000, 'Ravi': 40000}

Highest Salary Employee: Priya - 65000
Lowest Salary Employee: Ravi - 40000
Average Salary: 51000.0

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===== RESTART: C:/query processing/10.py =====
Marks: [75, 82, 91, 45, 67, 58, 39, 88, 95, 72]
Highest Mark: 95
Lowest Mark: 39
Average Mark: 71.2
Number of Students Passed: 8

>>>
```

```
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employees = {
    "John": 50000,
    "Ram": 45000,
    "Priya": 65000,
    "Anu": 55000,
    "Ravi": 40000
}

print("Employee Details:")
print(employees)

highest = max(employees, key=employees.get)
lowest = min(employees, key=employees.get)
average = sum(employees.values()) / len(employees)

print("\nHighest Salary Employee:", highest, "-", employees[highest])
print("Lowest Salary Employee:", lowest, "-", employees[lowest])
print("Average Salary:", average)

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0 Laptop 800 68000
1 Mouse 20 1700
2 Keyboard 35 2975
3 Monitor 200 17000
4 Printer 150 12750
5 Tablet 300 25500
6 Speaker 50 4250
7 Mobile 600 51000
8 SSD 80 6800
9 Camera 450 38250

>>>
===== RESTART: C:/query processing/10.py =====
Monthly Sales:
0 January 25000
1 February 30000
2 March 28000
3 April 35000
4 May 40000
5 June 38000
6 July 42000
7 August 39000
8 September 41000
9 October 45000
10 November 43000
11 December 47000

Total Sales: 453000
Average Sales: 37750.0
Maximum Sales: 47000
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===== RESTART: C:/query processing/10.py =====
Employee Details:
{'John': 50000, 'Ram': 45000, 'Priya': 65000, 'Anu': 55000, 'Ravi': 40000}

Highest Salary Employee: Priya - 65000
Lowest Salary Employee: Ravi - 40000
Average Salary: 51000.0

>>>
```

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10.py - C:/query processing/10.py (3.14.0)
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import pandas as pd

# Create the dataset
data = {
    "Month": ["January", "February", "March", "April", "May", "June",
              "July", "August", "September", "October", "November", "December"],
    "Sales": [25000, 30000, 28000, 35000, 40000, 38000,
              42000, 39000, 41000, 45000, 43000, 47000]
}

# Create DataFrame
df = pd.DataFrame(data)

print("Monthly Sales:")
print(df)

# Calculate Aggregation
print("\nTotal Sales:", df["Sales"].sum())
print("Average Sales:", df["Sales"].mean())
print("Maximum Sales:", df["Sales"].max())
print("Minimum Sales:", df["Sales"].min())

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>>> handle = open(
FileNotFoundError: [Errno 2] No such file or directory: 'product.csv'
>>>
===== RESTART: C:/query processing/10.py =====
product.csv created successfully.
>>>
===== RESTART: C:/query processing/10.py =====
Product Price_USD Price_INR
0 Laptop 800 68000
1 Mouse 20 1700
2 Keyboard 35 2975
3 Monitor 200 17000
4 Printer 150 12750
5 Tablet 300 25500
6 Speaker 50 4250
7 Mobile 600 51000
8 SSD 80 6800
9 Camera 450 38250

>>>
===== RESTART: C:/query processing/10.py =====
Monthly Sales:
0 January 25000
1 February 30000
2 March 28000
3 April 35000
4 May 40000
5 June 38000
6 July 42000
7 August 39000
8 September 41000
9 October 45000
10 November 43000
11 December 47000

Total Sales: 453000
Average Sales: 37750.0
Maximum Sales: 47000
Minimum Sales: 25000

>>>
```

Ln: 21 Col: 42

```
10.py - C:/query processing/10.py (3.14.0)
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import pandas as pd

data = {
    "Product": ["Laptop", "Mouse", "Keyboard", "Monitor", "Printer",
               "Tablet", "Speaker", "Mobile", "SSD", "Camera"],
    "Price_USD": [800, 20, 35, 200, 150, 300, 50, 600, 80, 450]
}

df = pd.DataFrame(data)

df["Price_INR"] = df["Price_USD"] * 85

print(df)
```

```
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>>> handle = open(
FileNotFoundError: [Errno 2] No such file or directory: 'product.csv'
>>>
===== RESTART: C:/query processing/10.py =====
Traceback (most recent call last):
  File "C:/query processing/10.py", line 4, in <module>
    df = pd.read_csv("product.csv")
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1026, in read_csv
    return _read(filepath_or_buffer, kwds)
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 620, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1620, in __init__
    self.engine = self._make_engine(f, self.engine)
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1880, in _make_engine
    self.handles = get_handle(
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/common.py", line 873, in get_handle
    handle = open(
FileNotFoundError: [Errno 2] No such file or directory: 'product.csv'
>>>
===== RESTART: C:/query processing/10.py =====
product.csv created successfully.
>>>
===== RESTART: C:/query processing/10.py =====

```

	Product	Price_USD	Price_INR
0	Laptop	800	68000
1	Mouse	20	1700
2	Keyboard	35	2975
3	Monitor	200	17000
4	Printer	150	12750
5	Tablet	300	25500
6	Speaker	50	4250
7	Mobile	600	51000
8	SSD	80	6800
9	Camera	450	38250

Ln: 83 Col: 33

```
10.py - C:/query processing/10.py (3.14.0)
File Edit Format Run Options Window Help

import pandas as pd

employees = pd.DataFrame({
    "EmpID": [101, 102, 103, 104, 105],
    "Name": ["John", "Ram", "Priya", "Anu", "Ravi"],
    "DepartmentID": [1, 2, 1, 3, 2]
})

departments = pd.DataFrame({
    "DepartmentID": [1, 2, 3],
    "DepartmentName": ["HR", "Finance", "IT"]
})

merged = pd.merge(employees, departments, on="DepartmentID")

print(merged)
```

```
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>>> File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1620, in __init__
self.engine = self._make_engine(f, self.engine)
File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1880, in _make_engine
self.handles = get_handle(
File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/common.py", line 873, in get_handle
handle = open(
FileNotFoundError: [Errno 2] No such file or directory: 'employees.csv'
>>>
===== RESTART: C:/query processing/10.py =====
File "C:/query processing/10.py", line 4, in <module>
employees = pd.read_csv("employees.csv")
File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1026, in read_csv
return _read(filepath_or_buffer, kwds)
File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 620, in _read
parser = TextFileReader(filepath_or_buffer, **kwds)
File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1620, in __init__
self.engine = self._make_engine(f, self.engine)
File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1880, in _make_engine
self.handles = get_handle(
File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/common.py", line 873, in get_handle
handle = open(
FileNotFoundError: [Errno 2] No such file or directory: 'employees.csv'
>>>
===== RESTART: C:/query processing/10.py =====

```

	EmpID	Name	DepartmentID	DepartmentName
0	101	John	1	HR
1	102	Ram	2	Finance
2	103	Priya	1	HR
3	104	Anu	3	IT
4	105	Ravi	2	Finance

Ln: 43 Col: 0

```
4.py - C:/query processing/col at1/4.py (3.14.0)
File Edit Format Run Options Window Help
import pandas as pd

# Create dataset
data = {
    "Product": ["Laptop", "Mouse", "Keyboard", "Monitor", "Printer",
               "Tablet", "Speaker", "Mobile", "SSD", "Camera",
               "Router", "Headset"],
    "Sales": [55000, 800, 1500, 12000, 7000,
              25000, 3500, 40000, 6000, 30000,
              4500, 2800]
}

df = pd.DataFrame(data)

# Sort by Sales in descending order
sorted_df = df.sort_values(by="Sales", ascending=False)
print("Top 10 Highest Selling Products:")
print(sorted_df.head(10))

Ln: 20 Col: 0
```

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Cleaned Dataset:
  Name  Marks
0  Arun  80.000000
1  Bala  81.666667
2  Charan 75.000000
3  Divya  90.000000
4  Esha  81.666667

>>>
===== RESTART: C:/query processing/col at1/3.py =====
Original Dataset:
  ID Name  City
0  101 John Chennai
1  102  Ram Madurai
2  101 John Chennai
3  103  Anu Salem

Duplicate Records:
  ID Name  City
2  101 John Chennai

Updated Dataset:
  ID Name  City
0  101 John Chennai
1  102  Ram Madurai
3  103  Anu Salem

>>>
===== RESTART: C:/query processing/col at1/4.py =====
Top 10 Highest Selling Products:
  Product Sales
0  Laptop 55000
7  Mobile 40000
9  Camera 30000
5  Tablet 25000
3  Monitor 12000
4  Printer 7000
8  SSD 6000
10 Router 4500
6  Speaker 3500
11 Headset 2800

Ln: 68 Col: 0
```

```
3.py - C:/query processing/col at1/3.py (3.14.0)
File Edit Format Run Options Window Help
import pandas as pd

# Create dataset
data = {
    "ID": [101, 102, 101, 103],
    "Name": ["John", "Ram", "John", "Anu"],
    "City": ["Chennai", "Madurai", "Chennai", "Salem"]
}

df = pd.DataFrame(data)
print("Original Dataset:")
print(df)

print("\nDuplicate Records:")
print(df[df.duplicated()])

# Remove duplicates
df = df.drop_duplicates()
print("\nUpdated Dataset:")
print(df)

Ln: 23 Col: 0
```

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
File "C:/Users/A9194/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/_io/_common.py", line 973, in get_handle
    handle = open(
FileNotFoundError: [Errno 2] No such file or directory: 'student_marks.csv'

>>>
===== RESTART: C:/query processing/col at1/2.py =====
Missing Values:
  Name Marks
0 False False
1 False True
2 False False
3 False False
4 False True

Cleaned Dataset:
  Name  Marks
0  Arun  80.000000
1  Bala  81.666667
2  Charan 75.000000
3  Divya  90.000000
4  Esha  81.666667

>>>
===== RESTART: C:/query processing/col at1/3.py =====
Original Dataset:
  ID Name  City
0  101 John Chennai
1  102  Ram Madurai
2  101 John Chennai
3  103  Anu Salem

Duplicate Records:
  ID Name  City
2  101 John Chennai

Updated Dataset:
  ID Name  City
0  101 John Chennai
1  102  Ram Madurai
3  103  Anu Salem

Ln: 54 Col: 0
```

```
2.py - C:/query processing/col at1/2.py (3.14.0)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

data = {
    "Name": ["Arun", "Bala", "Charan", "Divya", "Esha"],
    "Marks": [80, np.nan, 75, 90, np.nan]
}

df = pd.DataFrame(data)

print("Missing Values:")
print(df.isnull())

df["Marks"] = df["Marks"].fillna(df["Marks"].mean())

print("\nCleared Dataset:")
print(df)
```

Ln: 18 Col: 0

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/query processing/col at1/2.py =====
Traceback (most recent call last):
  File "C:/query processing/col at1/2.py", line 4, in <module>
    df = pd.read_csv("student_marks.csv")
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1026, in read_csv
    return read(filepath_or_buffer, kwds)
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 620, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1620, in _init
    self.engine = self._make_engine(f, self.engine)
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/parsers/readers.py", line 1880, in _make_engine
    self.handles = get_handle(
  File "C:/Users/91944/AppData/Local/Programs/Python/Python314/Lib/site-packages/pandas/io/common.py", line 873, in get_handle
    handle = open(
FileNotFoundError: [Errno 2] No such file or directory: 'student_marks.csv'
>>>
===== RESTART: C:/query processing/col at1/2.py =====
Missing Values:
   Name  Marks
0  False  False
1  False   True
2  False  False
3  False  False
4  False   True

Cleared Dataset:
   Name  Marks
0  Arun  80.000000
1  Bala  81.666667
2  Charan  75.000000
3  Divya  90.000000
4  Esha  81.666667

Ln: 23 Col: 0
```